All right, good afternoon. Go ahead and start now. Recap from last time. So we've been talking about the Bayesian framework for doing statistical inference. We talked about how to get a posterior distribution. But then we wanted to think about how to get a point estimate in the Bayesian framework. And the way we do this is with loss functions. So we just went back over what a point estimate is or a point estimator. It's kind of our single best guess for the unknown parameter value. Right. But in the Bayesian framework, we have an entire distribution that expresses our knowledge about the parameter at any given state of time, right? And so this entire distribution tells us everything, but it is not a single best guess. We can use it to get a single best guess to get a point estimate, but by itself it is not. And so we do this through loss functions, right? And so we set a loss function is just some real valued function of two variables. In our case, these will be theta and a, theta being the parameter, of interest and A being our estimate or our guess, right? And so we can choose a loss function based on kind of our own preferences, our own tolerances, on how we estimate this thing, right? Do we want to penalize big deviations more or less? And so that's the idea. We can choose a loss function and that loss function will help us or tell us what our single best guess is given our posterior distribution, our state of knowledge about the parameter. And so we call this a base estimator, a Bayes estimator, given some loss function, L of Theta and A, right, the base estimator is just the one that minimizes this posterior expected loss.

And we said we have a distribution. We don't know the loss value itself because we don't know the parameter. We have a distribution for the parameter. So we can calculate expectations of anything involving that parameter or in particular, a loss function involving that parameter.

And the posterior expected loss is what we'd like to minimized. We'd like that to be as small as possible. On average, can we make our loss function as small as possible? And if we do that, that's called the base estimator. That's our single best guess or our point estimate under the Bayesian framework for a given loss function. So it depends on two things. It depends on our posterior distribution. And it depends on the loss function that we chose. We looked at two main loss functions here. So we talked about squared error loss, right? Just the difference between the parameter and the estimate squared. And we looked at absolute. error loss, which is the absolute value of that difference. And we talked about how this first one penalizes big deviations more than the second one. But there's, of course, other types of loss functions.

You could have asymmetric loss functions, any number of loss functions. These two are very, very common. So this is what we worked with here.

We said four squared error loss. So whenever we're using the squared error loss function, the base estimate, or the one that's going to minimize that posterior expected loss is the posterior mean, the mean of our posterior distribution or the expectation of the parameter theta given the data X.

So anytime we use squared error loss, the Bayes estimator is going to be the posterior mean. Now what the posterior mean is will be different depending on what our posterior distribution is, but it's always going to be the posterior mean. We did some examples there. And then for the case of absolute error loss, the Bayes estimator is the posterior. median. Again, what that median is might be different depending on what your posterior distribution is, but any time you're using absolute error loss, you know you can find the posterior median and that is your Bayes estimator, or the one that minimizes the posterior expected loss, posterior expected absolute error loss in this case.

Okay, so we're going to transition away from the Bayesian framework now.

Talk about maximum likelihood estimation.

Okay, so we're moving away from the Bayesian framework. We'll talk about it again more in the future, but for now, we're kind of transitioning. We're going

to talk about point estimation in the frequentist framework. So a couple things to keep in mind. We're switching mindsets here. We're no longer using a prior distribution. That won't be part of this at all. We're no longer treating the parameter theta as a random variable. Instead, we're treating it as some fixed but unknown quantity. It's just some constant value that's not random, but we don't know what it is. We'd like to estimate it. That's the frequentist framework for probability or thinking about a parameter. And we're going to talk about the maximum likelihood estimator, which is the main way that Frequantists estimate a parameter or get a point estimate for a parameter. So the maximum likelihood estimator is just given here. We're going to use this notation, Theta hat. The hat's just going to tell us that it's an estimate. So Theta hat is our estimate, our best guess. And it's going to be a function of the data. So we're going to take in the data and output some point estimate, some best guess. And the way we do this is this says the argument of theta. that maximizes the likelihood function, provided it exists. So the argument of theta that maximizes the likelihood function, what does that mean? Well, we've talked about the likelihood function. It's some function of the generic parameter theta. And we could plot this. You did this for one of your homework problems. Maybe it looks something like this. Maybe this is your likelihood function. So on the x-axis, we have the parameter theta. On the y-axis, we have the likelihood. We don't actually really care about what the actual the actual magnitude of the likelihood is, it's all relative, right? Meaning this value is, in some sense, more likely than this value. The inputs associate with those values, this theta. That's all the likelihood function tells us. And so here we want to, we want the argument of theta that maximizes the likelihood. So what does that mean? We're going to find the maximum of this likelihood function, right here. If we draw these red dashed lines on the y-axis, this is the max. likelihood.

It's the value of the likelihood at its maximum.

On the x-axis, we have theta that gives us that maximum likelihood. And the theta that gives us that maximum likelihood is what we really care about. That's theta hat. That's our maximum likelihood estimates.

The argument of theta or the input of theta that gives us our maximum.

So that's all that this equation is saying. It's the argument of theta, the input in this parameter space, such that the result is the maximum value for the likelihood. That's on. our point estimate or a maximum likelihood estimator. And just like we've talked about before, right, after we observed data, so capital X becoming some observed value little X, then we call that the maximum likelihood estimate of data. So the maximum likelihood estimator is the kind of abstract concept or rule for getting a point estimate under a specific likelihood function, and the maximum likelihood estimate is the specific estimate we get from an observed data set. So we'll use MLE for short for maximum likelihood estimate or maximum likelihood estimator. The context is usually clear which. But hopefully a fairly intuitive framework here, right, pretty simple conceptually. We have a likelihood function. We don't really care what the value of the likelihood is at the maximum, but we care about the value of theta that gives us that maximum. We want that value to be our point estimate. And that's kind of all there is to it. That's maximum likelihood estimation. Now we actually need to do that. So I'm going to talk about kind of a general procedure that works in a lot of cases, not necessarily every case. But so logarithms, log transformation is monotonically increasing. So the argument or the theta that gives us the maximum likelihood will also give us the maximum log likelihood. So if we took the log transformation of this function, right, the theta that maximizes its original function would be the same theta that maximizes the log of this function. Okay, why do we care about that? Well, it can often make computation much easier. So we've seen when we write down a likelihood function, it's a product of a bunch of terms. Those products can be messy. They can, you know, if you take a product of a bunch of really small values, you can get underflow. So we might not want to deal with that product. When we take the log transformation, products become sums. Sums can be a lot easier to work with. You might have also noticed that a lot of probability distributions, PDFs or PMFs,

right, They have E to some power. When we take the log of that, we just isolate the power. So there can be computational or mathematical reasons why we might want to work with the log likelihood instead. But the point is that maximizing the likelihood and maximizing the log likelihood are the same. Doing either gives us the same theta hat. So let's do the one that's easier to work with, work with, which in many cases will be the log likelihood. So that's going to be convenient. And so this is the idea. We're going to first compute the log likelihood function. So this is the notation I will use for log likelihood, this lowercase cursive L of theta given X. It's just the log of the likelihood function. So we already know how to find the likelihood function to take the log of that. Now, if we want to optimize something, if we think back to calculus, we have one way to do this. If we want to identify this maximum, what's the derivative of this function with respect to theta at that maximum? zero, right?

So we could take the derivative of this function and set it equal to zero. Try to optimize it that way. Or if we're working with the log likelihood, take the derivative of the log likelihood set it equal to zero. And that's exactly what we're going to do. So we compute the log likelihood, that's step one. Then we differentiate that log likelihood with respect to theta, and we're going to set that equal to zero. So D by D theta of the log likelihood equal to zero.

Okay. And then we're just going to solve that expression for theta find find this specific theta here that gives us that zero derivative or potential maximum and we're going to call that our candidate m le or theta hat now the derivative could also be zero at a minimum we don't want that we want a maximum right so we always want to check the second derivative make sure that that's less than zero make sure it's a concave function at that point right so that we're getting a maximum and not a minimum for a lot of basic models like we work within this class. This will be concave over the whole parameter space. But for more complex models, it might not be. Yeah, question over here. By log being monotonically increasing, I mean that if Theta 1 gives a higher likelihood than Theta 2, then it will also give a higher log likelihood than Theta 2, right? So if you plot the log function, the log function looks like that. It is a monotonic function. It doesn't mean that the log likelihood, is monotonic. But the log function is monotonic, which tells us that for any Theta 1 with higher likelihood than Theta 2, Theta 1 will have a higher log likelihood. It doesn't mean that the likelihood itself is monotonic or that the log likelihood is monotonic. It could be, it could not be.

Okay, so that's the general strategy. Again, this is not the definition of the maximum likelihood estimator.

This is the definition. But in many cases, this will give us or give us a way to solve for the maxim likelihood. We'll look at some examples where it doesn't, though. Yeah, question. It's a characteristic of the log function, yeah, which is separate from our parameter, our model, anything. It's just the log function. If you plot the log function, it's monotonic increase. All that means is that X1, X2, right? X2 is greater than X1, so log of X2 is greater than log of X1. That's all I'm saying, and you see that there, right? Yep, just for convenience. You don't have to use the log transformation here. you're welcome to solve it on the original likelihood. Often easier, almost always easier, to take the log transformation, but you certainly don't have to. So that's really it conceptually. We have a definition for the maximum likelihood estimator, and we have a routine that will give us that maximum likelihood estimator in many cases. And so now I just want to go through examples and show you in some cases where we compute this maximum likelihood estimate. So let's say we have a random sample of size 3, coming from an exponential distribution with parameter theta. So the individual data points, the X's, follow an exponential distribution. It's a random sample. So they are IID from that distribution, independent and identically distributed from that exponential distribution. And we observe some data points. X1 is 3, X2 is 1.5, and X3 is 2.1. We're asked to compute the MLE of theta. Okay? The MLE is the maximum likelihood. So we know we're going to need our likelihood function. So let's go ahead and start there. The likelihood function is a product of the individual PDFs or PMFs of the data points. So we

need the PDF in this case of an individual data point coming from this exponential distribution. So f of X given theta, the distribution of an individual data point is exponential. So it has PDF theta, E to the negative theta X greater than equal zero. Okay, that's one data point.

Let's start by writing down our likelihood function now, L, theta, given the data, X.

And we know that the likelihood function is the product of the individual probability density functions in this case.

Or the product, I equals 1, n, xI, given theta, which for this specific case, is the product from I equals 1, 3 of theta, e to the negative theta xI. So we have product of three of these theta terms, theta to the third. And then inside the exponent, we have each of these negative theta to the xI, which simplifies to e to the negative theta sum of the xI. This is the likelihood function. I'm just going to say theta greater than zero here. This is the likelihood function, but I'm going to take the log of this. I'm going to get the log likelihood function. It's going to be easier to work with. So I denote that little L theta given X, which is just. just the log of that likelihood function that we just wrote down this expression here.

So we have three log theta minus theta the sum of xI, sum from i equals one to three. Okay, this is our log likelihood function, the log of the likelihood.

Okay, that was step one.

Step two was to differentiate this log likelihood with respect to our parameter theta. So we want d by theta of this log likelihood function that we just wrote down. D by the theta of 3 log theta minus theta times the sum. And this becomes 3 over theta minus the sum of our X's. We got our likelihood function, we took the log of it, and we differentiated that, and that gives us a candidate point. We want to find the maximum, so we want this candidate point by setting it to 0. We're going to set this expression, the derivative of the log likelihood function, to be zero. Zero equals three over candidate Theta hat that we're going to solve for minus the sum of our data, our X values. Set this to zero, find where the derivative is zero, and solve for Theta, and that gives us a candidate. Okay, so solving for Theta or Theta hat, we get Theta hat equals three over sum of our X's, which in this case is three over 6.6. Now I just want to double check, but this is a maximum and not a minimum. Step four, the second derivative of this log likelihood function. Or just differentiate one more time, D by a theta of 3 over theta minus the sum of our X values, which is negative 3 to theta squared. So is this a maximum or is this a minimum? It's a maximum. It's negative, right? 3 is positive. Theta is the parameter of an exponential distribution, which must, be positive.

That's a parameter restriction, right? So this entire thing is a negative number, less than zero.

Questions on this example? Yeah. Theta, instead of theta the third, you're saying just theta.

So you'll have negative theta times X1 plus negative theta times X2, plus negative theta times X3. So you can factor out the negative theta times the sum of the X's. Does that make sense?

Yeah, question. Yeah, if you didn't have the three observed data points, right? So say X1, X2, and X3 are still random, not observed, then this, this piece here would be your maximum likelihood estimator.

This is our maximum likelihood estimate. So that's exactly what we said in the definition. The estimator is the general rule for capital X. When you plug in little X, that's a maximum likelihood estimate. So abstract concept, general rule for calculating it, specific point estimate given data.

Let's do some more examples.

So now we have data observations x1 through xn again this is a random sample now from the bernoulli distribution 0.1 coin flips and this bernoulli distribution has parameter p it's the maximum likelihood estimator for p so we don't have observed data here we're just finding that general rule abstract concept here right what's the general rule for estimating p in this setup bernoulli data and well the maximum likelihood estimator we know we need the likelihood function the likelihood function involves the individual PDFs or PMFs of the data. In this case, our data is Bernoulli discrete as a PMF. So what is that? F of X given p, p to the X times 1 minus p to the 1 minus X, and X can be either zero or one. Those are the allowable values for X. This is the distribution for a single observation, but we can use it to get our likelihood function. So just like before, step one, I'm going to write down the likelihood function. L, P, given the data, X in this case. As always, it's just the product from I equals 1 to n of the individual PMFs in this case, F of xI given p. So we have the product, I equals 1 to n of these functions, these Bernoulli probability mass function. P to the xI times 1 minus p, the 1 minus xI, which simplifies to p to the sum of the x's times 1 minus p to the n minus the sum of the x is. And then I'll just clarify, but p is between 0 and 1. So this is our likelihood function. If you wanted to, you could optimize this. I'm going to instead work with the log likelihood function. Little L, p given x. Just the log, this likelihood function that we just wrote down. We just take the log of that. And so that is the sum of x times log p plus and minus the sum of xI. times the log of 1 minus p. Started with our likelihood function. Took the log of that because it's going to be easier to work with. Step 2 is what? Yeah, we're going to take the derivative of this log likelihood function and set it to 0. We want d by dp in this case of L, p given it. Okay, so the derivative of the first term, sum of xI or p, derivative of the second term, negative n minus the sum of the x is 1 minus p. Started with our likelihood function, took the log, took the derivative of that with respect to the parameter, and now we're going to set that equal to zero and solve for a candidate value. 0 be equal to sum of x, p hat, minus n minus the sum of x over one minus okay. And there we can get sum of xx. The hat is equal to n minus the sum of x i. we get one minus the sum of x i full two p hat sum of x there we have sum of x i the p hat is n and lastly solving for p hat we get some of our x's some of our data over n what is this sum of x over n yeah it's the average the sample mean in general or for the specific case of binary data the sample proportion if we think about this as coin flips right if we flip 10 coins and eight of them come up heads the sample proportion is 8 out of 10.8, it's the maximum likelihood estimate. This problem, is that normal? Not necessarily, it is frequently the sample mean. Sometimes it's the inverse of the sample mean, like in this exponential case. But there's many cases where it's something else as well. But yeah, the sample mean does come up a lot, especially when you're estimating things like the mean. It very often is the sample mean. Yeah, question back here. So I'm just saying, right, we use hat to denote the estimator. And since I'm solving for our candidate estimator, I'm just putting a hat on it. Yeah, it's just the notation I'm using to clarify that it's the potential estimator. If you wrote P there and you still solved and then somehow tell me that your estimate of P is this, P itself is not that. We don't know P. We don't know the true value of P, but we have an estimate of P. So if you somehow tell me that your estimate of P is this maximum likelihood estimate, that's fine and just make that clear. Yeah, I use the hat for that notation. But if you wrote down P is equal to that, right, that's not true. We don't know a P. This is just our best guess for it. Yeah, question. Yeah, if you have, so what we showed here is that if you have a random sample from the Bernoulli distribution, the maximum likelihood estimator for that parameter P is always the sample mean. Or once we show that the second derivative is negative, we'll have confirmed that. But yeah, anytime you have this setup, a Bernoulli distribution, random sample from it, we can use the sample mean as a maximum likelihood estimate for the parameter P. You could plug in the individual data points X. you could plug in your sample size N for a specific observed data set. Yeah, here we showed basically that it is we did it for a specific data, but before we plugged in the data points. Not necessarily, right, because you could

always derive it from the fact that you're just maximizing the likelihood. Any number of cases that could come up, right? So you should always be familiar with how to get to that rather than just memorizing what the answer is. But if you're in the real world and you happen to collect Bernoulli data from a random sample and you want a maximum likelihood estimate and you remember, sure, go ahead and plug it in. Is there one more question? And before we move on, step four, we want to confirm that this is indeed a maximum. So we want to take the second derivative of this log likelihood function or differentiate one more time from the expression we had sum of x over p minus n minus the sum of x minus p. and when we differentiate again we get negative sum of x over p squared minus n minus the sum of x over one minus p squared and since p is between zero and one this will be negative so we have found a maximum not a minimum another example data x1 through xn that form a random sample now from the normal distribution and we don't know the mean of that distribution mu but we do know the variance sigma squared So one parameter, one unknown, the mean mean. Let's find the maximum likelihood estimator for that mean. Okay. Same idea as before. We want to maximize the likelihood function. We need to write down our likelihood function. The likelihood function is going to be a product of the individual PDFs in this case. Because we have a continuous distribution for the data, normal distribution. And that PDF, f of x given mu, is just the PDF of a normal distribution, which you can look up.

It's one over square root, two, pie. sigma squared, e the negative x minus mu squared over two in a squared.

This is the distribution or the PDF of a single data point, a single normal random variable. But we don't have a single one, we have n of them from this random sample. So that's going to tell us our likelihood function. Step one, write the likelihood function, L of mu given the data x.

It's just going to be the product of these n probability density functions pdfs so the product from i equals 1 to n of little f of x i given mu the product equals 1 to n 1 over the square root of 2 pi sigma squared e the negative x i minus mu squared 2 sigma squared okay this term does not involve the parameter mu we're going to have n of these terms in the product 1 over square root of 2 pi sigma squared 2 the nth power and again inside the exponent these terms will be additive so we will have e to the negative sum one to n x i minus mu squared over to sigma squared this is our likelihood function for the parameter mu and there's no restrictions on mu it can be any real number okay i don't want to work with this i want to work with the log likelihood so i'm going to write down l a little of mu given the data x and take the log of its likelihood expression that we just found The log of that, minus n 2, log 2 pi, sigma squared, summation, 1 to n, xI minus mu squared over 2. Now notice, this second term contains the parameter mu. This first term does not. This first term is constant with respect to mu. So when we differentiate, this is going to go away. Cross that out, cross that out, it's a constant. You can get rid of the whole thing, yeah. Because it doesn't involve mu, so when we differentiate, it goes away. So just keep in mind that's a constant. When we differentiate, it's only going to involve this second piece here.

So we're going to work with the log likelihood function. We're going to differentiate that d by d mu x, which is just the derivative of this second term.

And that is the sum x i minus mu over sigma squared.

And now we want to take that derivative and set it equal to zero and solve for mu.

And that gives us our candidate zero is equal to the summation x i minus mu hat over sigma squared 0 is equal to summation x's minus n mu hat. Multiplying by sigma squared, we have the sum of the x's. The sum of n of these mu hat terms is n mu hat. And then we solve candidate mu hat going to be sum of the x's over n. So we got our sample mean again. It makes intuitive sense. We're estimating a

mean parameter. The maximum likelihood estimate of that mean parameter is the sample mean. Just like before, the mean of the Bernoulli distribution is P. Our maximum likelihood estimate for that parameter p is the sample mean. Before we forget, let's differentiate one more time. M squared of our log likelihood function, L of mu, given X, the derivative with respect to mu, summation xI minus mu over sigma squared. And this is negative n over sigma squared. Again, it's negative less than zero. So this is concave, we have a maximum and not a minimum. One more kind of standard example before we move on to the weird ones. Where the double derivative is like zero and therefore we have to like do you guys that is in the MLE in that case? Potentially. Yeah, always check. Yeah, you could have a likelihood function like that and this, this, and this I'll have derivative zero. Could have say this is your parameter space. You could have something like this. This is the only point that has derivative zero. It's not the maximum. In fact, this point on the edge is the maximum. maximum. You would never get that point by taking the derivative of the likelihood or the log likelihood and saying it to zero, right? So this is what I would call an edge case, but these do come up, and we'll talk about those in a moment.

Yeah, question. Yeah, so after this next example, we'll talk about some of those cases.

Okay, so one more standard example.

The number of failures in a computer system in one day follows a Poisson distribution with parameter lambda. The distribution of a data point is Poisson, unknown parameter Lambda. We observe failures on seven consecutive days. These are those failure counts. Let's find the MLE maximum likelihood estimate for this parameter. The same idea. We need our likelihood function. PMF of the data in this case of an individual data point helps us get there. Lambda is the Poison probability mass function. Lambda, the X power, E negative lambda. Step one, write down our likelihood function, capital L. of Lambda, the parameter, the data.

I'm a sample, so it's the product, the N, of these individual probability mass functions.

F of XI, these are poisson, so we know this is going to be the product, the N of lambda, the XI, E, the negative lambda, XI, factorial. We're collecting terms and simplifying.

We get lambda the sum of XI power, E negative N lambda, times this product of these one over x factorial terms.

Notice this involves lambda the parameter.

This involves lambda the parameter.

This does not. This is constant with respect to the parameter.

Okay, I want to work with the log likelihood.

Little L of lambda and x. I'm going to take the log of that likelihood function and we get the summation of xI.

There's log lambda. This n lambda. And I'm just going to say plus constant.

I really don't care about this term here. It doesn't involve lambda constant with respect to lambda. It's going to disappear when we differentiate. Okay, so we have our log likelihood function now.

Step two, we're going to differentiate it. Set that to zero.

D by D lambda, this log likelihood function, middle L of lambda and x. And we have sum of our data, sum of x over lambda minus n.

Just set that full to zero. Solve for lambda.

Zero is equal to summation of our data. And solving for lambda or lambda hat, we get sample mean. of x's over n. And because we were given specific data points, right, sample size of seven with these distributions here, we can go ahead and plug those in. In this case, 22 over seven is our maximum likelihood estimate of the parameter lambda, as long as we differentiate one more time and confirm that it's a minimum. Yeah. Some of x over the lambda equal minus 10. But then next you wrote lambda equal sum of x over n So I wouldn't be like n is equal to the sum of x over? So I added n to both sides. Multiply both sides by lambda hat and divided by both sides by n. So I'm just solving for a lambda hat there. The last step, just verify. This is concave that we have a minimum. Second derivative of our log likelihood function. L of lambda x. Or just differentiate one more time. D by d lambda. The sum of x over lambda minus n. And this is the negative. summation of x over lambda squared. Sum of x is positive. Lambda is positive. So this expression is negative. So we have a maximum rather than a minimum. So we had four nice examples where our little recipe, this procedure works. These questions on this procedure, yeah. Yeah, if you have individual data points, you have a sample size, you have values for X, calculate random variables X equals some value. If you don't calculate the general rule, the maximum likelihood estimator, not the maximum likelihood estimate, or both if I ask you to do both. Okay, so let's look at when this doesn't work.

Sometimes you might not have a local maximum within the parameter space. Right, I drew a little plot here where we don't have that.

Sometimes you might have a local maximum in the parameter space, but you might still have a global maximum on the edge, edge of the parameter space like we saw in that graph.

When you have situations like this, that procedure, that rule that I gave you is not going to work because it's going to identify those interior points where the derivative is zero. It's not going to identify edge cases like this.

I encourage you in general to check the boundary points, the edge points, if there are parameter restrictions. If you have something like this normal distribution here, there are no edges, there's no boundaries, so there's nothing to worry about there. But if you have edges or boundaries, I encourage you to check the end points. But where you often see this as a problem is when the support of the data depends on the parameter.

The parameter changes than the allowable values that my data take change. If that's the case, you'll often run into these edge cases. So let's look at a couple examples where that is the case.

Let's say we have data X1 through XN and it's a random sample. Now from the uniform distribution on this interval, the interval zero to theta.

So the lower bound zero of this uniform distribution is known.

The upper bound is some positive value theta that's unknown.

That's the parameter. So we want to find the MLE of this parameter theta. And so here's a case where the support of the data or the allowable values that X can take depends on the parameter, right? If theta is three, we can get values of X between two and three. But if theta is two, we can't get values of X. The support of the data depends on the parameter. That wasn't the case of the other examples we looked at. In the normal example, the data, X, can be any real number, regardless of what the mean Mew is.

In the Poisson example, the data X can be any non-megavinteger, regardless of the parameter, might make some values more likely than another, but it doesn't restrict the allowable values of X. Here we have restrictions on what values X

can take based on the parameter. Yeah, not necessarily, but I'm just saying, like, hint, hint, they often show up when you have a situation like this where the support of the data depends on the parameter. In general, it's good to check the boundaries if you have boundaries, but very often this is the type of situation where those show up.

So let's go ahead and find the maximum likelihood estimate or estimator, I guess, in this case.

Same idea. We're going to work straight from the definition of maximum likelihood.

We know we need to write down our likelihood function and maximize that function, whatever it is. And so we need the distribution of our data, F of X given theta.

If we have a uniform distribution between zero and theta. It's a continuous random variable. What is this PDF equal to? So the beta distribution is always restricted to be between zero and one, right? So the uniform zero to one is a special case of the beta, but the uniform zero to theta is not.

Okay, so it's uniform.

So this function should be constant. It should be the same for any value of theta, right?

What value should it be? Yeah. Yeah, it's one over theta.

Why? It's a probability density function, so it has to integrate to one, right? So if we have zero, theta, we know it's some constant value over that interval.

Say this is the height, H.

It's uniform distribution, so it's constant. We know it's going to be a line. It's going to be the same value H everywhere, and it's a probability distribution, so it has to integrate to one. What's the integral from zero to theta of this?

It's Theta H. So what value of H integrates to one? One over theta. So we know that this is going to be 1 over theta. X, usual X, X has to be between zero and theta. This is the probability density function for a single data point, a single X. But we can use this to get our likelihood function. So our likelihood function, L of theta, given X, straight from the definition of a likelihood function is the product, I equals 1, the N of the PDFs, F of X of I, given theta. Product over theta, and I'm going to use the cater function for this. support of our data depends on the parameter, so I'm going to use an indicator function to reflect that. So times an indicator, x is less than theta. You don't need to specify that it's greater than zero because zero is known.

Okay, so I have a product and indicator functions here. Each one is saying return one, if x i is less than or equal to theta, return zero otherwise. Can we simplify that at all?

Is there a single data point in the dataset that if this is true for that data point, it's true for all of the others? Data point in the data set, the X is? The last one, or is there a single X in the data set such that if this indicator is true for that point, then it must be true for every other point. If that X is less than equal to theta, then every single other X has to be the largest one, the maximum, right? So instead of this product of N indicators, we could just have the indicator for the maximum. That's a way to simplify this. I'm going to write this now as one over, over theta to the nth power times the indicator, just that the biggest one, the maximum xI is less than equal to theta. If the biggest one is less than equal to theta, then all the rest are two. So we can write it with one indicator function. Say that again. I are the individual data

points. N is the sample size. So here we have n because we have one of these terms for all n individuals. But here we're saying the maximum over all i's of the xIs. Just the biggest, the biggest X that we observe in the dataset. So this is the likelihood function. This is what we want to maximize. Notice theta is positive. So as theta gets bigger, this expression gets smaller. Theta gets bigger, one over theta to the end gets smaller. We don't want it to get smaller. We want to maximize it, we want it to get bigger, right? So we want smaller theta, not bigger theta. The smallest value of theta that we can take. Zero? What would happen to this indicator function if theta was zero? Yeah, equal to the maximum. This indicator tells us that theta has to be greater than or equal to the maximum. So the smallest value that theta can take is the maximum. Maximize this likelihood function. What did we say here? We said this is a decreasing function. Decreasing function of the parameter theta. By that, I mean 1 over theta 1 to the n is greater than 1 over theta 2 to the n if theta 1 is less than theta 2. Smaller values of theta give bigger values of the expression. decreasing function of theta.

This is a decreasing function of theta. We want to maximize it. So to maximize this, theta hat, MLE, must be as small as possible.

But it must be as small as possible without violating that indicator function. Otherwise, the whole thing returns zero. It has to do that while still satisfying constraint that we have, constraint that the maximum X value, maximum XI less than or equal to theta half.

So then theta hat, the maximum of X, X, Y, through XN.

So we didn't use calculus here, right? We just used this indicator function and the fact that that's a decreasing function of theta. We didn't take the derivative, work straight from the definition of the likelihood, and the fact that we're maximizing it.

Look at one more edge case example. Yeah, question.

You said it has to be, the data X has to be between zero and theta.

Why couldn't theta be twice the maximum?

Right? You have to use the fact that it's a decreasing function. We needed that so that we were forced to make theta as small as possible. If we didn't, we didn't didn't have that, we could pick any large value of theta, right? But the fact that the decreasing function forced us to make theta small, and the smallest we could make it was the max. But we only got that decreasing function by writing out the likelihood here. We had to look at, inspect what that function was.

The smallest possible value of X is zero.

Yeah. We're conditioning on the data. So if you had a data point that was five, theta can't be zero, right?

In fact, it can't be any value lower than any of your data points, meaning it can't be any value lower than the maximum data point. That's kind of the argument. That is the MLE. Yeah, I will circle that. That is the MLE of a random sample from a uniform distribution on zero to theta. This is the PDF of a single data point, one data point. It is not the likelihood function. We're interested in the likelihood function. This is the likelihood function, but this is not this. This function, our likelihood function, looks more like this. It's a decreasing function. So as theta gets bigger, the likelihood gets smaller until you get below the maximum, and then it drops to zero because the indicator function returns zero. So this is what our likelihood function looks like, not this. Does that make sense? Let's do one more edge case example. So a random variable X is modeled using the following PDF. So I don't have a name for this distribution. I'm just giving you the PDF now. It's not one of our known

distributions. We can still work with it, though. As long as we have a PDF we can work with. Let's say we have a random sample from this distribution. What's the MLA of theta? And you'll notice again, the support of the data X depends on this parameter of theta. The value of theta determines which values of X are allowable. Just like in the last example, although we have a little bit more complex probability density function here. The same principles are going to apply. We want to write down our likelihood function and we want to maximize that function, whatever it may be. So let's write down our likelihood function, L of theta, the data X. We know it's going to be product of individual probability density functions because we have a random sample, product of f of xI and theta, and we're given this function F.

It's given as part of the proper product. I equals 1, n 2, theta squared, xI indicator that xI is less than equal to theta.

Okay, let's simplify as much as possible.

We have n of these terms, 2 over theta squared, 2 over theta squared to the nth power. product, these xI terms.

And just like before, let's simplify this. We know if every single X is less than equal to theta, right? That's the same as saying the maximum of those has to be less than or equal to theta. So we can write this just with the single indicator maximum X less than or equal to theta.

This is constant with respect to theta. So this isn't going to help us.

We can ignore it. We know theta has to be greater than equal to the maximum. And this piece changes with theta.

But again, it's a deep. decreasing function of theta.

Theta gets bigger, one over Theta squared gets smaller, and the whole expression gets smaller. This piece is constant, so it doesn't change as we change theta.

Again, this is a decreasing function of theta.

So what's the MLE? It's the maximum.

It's a decreasing function of theta, meaning we'd like to make theta as small as possible because we want to maximize the likelihood, not minimize it. Smallest value of theta we can take, given by this indicator function here, is the maximum value of A. It's a maximum observed x value. So again, Theta hat, the maximum of our observed data, our random sample. And again, we're just saying that the likelihood function, if we have theta on the x-axis, the likelihood on the y-axis, let's say this point here, the maximum x-I. We're just saying the likelihood function looks something like this. It's a decreasing function, so it gets smaller and smaller as theta gets bigger. But because of that indicator function, it drops to, to zero for any value of theta less than the maximum. You would have an indicator function where, so you have a product of indicators where each one, each X has to be greater than or equal to theta. And so you could write that indicator function as the minimum X has to be greater and equal to theta. So you could work with the minimum. That's what I'll say. It could, yeah. Vitis value you observed would be the most likely estimate? Or would you ever want to? Yes, for like what the maximum value yeah you're getting at an interesting question there so right we collect these end data points we know theta has to be greater than them if we use the maximum of those xes as our estimate is that on average equal to the true value is that kind of what you're asking you think there could be an x value yeah we better think the maximum to be rather than what your actual yes in this case right i'll go to this uniform example in this case for a finite sample size the maximum x value is probably not theta. Theta is probably some value above the maximum, right? Some value past that. Now, as you collect more and more and more data, that maximum should get closer and closer and closer to theta, right? Does that make sense?

So the question you're asking is related to bias.

Is it a biased estimator? In this case, the answer is yes. And consistency, meaning as our sample size gets really large, goes to infinity, do we sort of converge to the right answer. And in this case, yes as well. Are you willing to have a biased estimator, though?

That's something to consider. A lot of people would say no. But great question. This stuff will talk about a little bit later in the course. So we will get to that. And I think we actually come back to the same example when we do talk about showing them it's a biased estimator.

Yeah, you could put a prior on theta, right? And then you're going to get some posterior mean that's not the maximum X. It's going to pull it that way. That is a solution. Another solution is to just use an estimate that's not the maximum likelihood. The maximum likelihood is by far the most common frequentest estimator, but it's not necessarily the only one. So you could try to find another estimator, has better properties that you care about. But yeah, MLE is kind of the main approach that people use. So they're like just two different ways kind of get at the same thing, which is like some sort of estimate, final estimate for data? Well, if you use a, Bayes estimator, you get a point estimate, and this gives you a point estimate. So a Bayes estimator and a maximum likelihood estimator are both point estimates. The Bayesian approach in general gives you a full posterior distribution, so you get a little more than just a point estimate there. You can do other things with it, but you can use it to construct a point estimator.

So far, we've only seen how to get a point estimator for the maximum likelihood estimate. We'll talk about other things we can do with it later. There is more possibilities.

We can study, like, say our MLE is the sample mean. The sample mean is itself random. It involves random data. So we could study the random properties of that estimator. So there's more we could do with it. We haven't talked about that yet. We will. But for now, it's just a point estimate.

So basically, like, maximum likelihood estimate is, like, mathematically is kind of the opposite of, like, the point estimate from a agent estimator. Because here you're taking, like, the maximum of the likelihood function, whereas with that, you would take the minimum of the loss function, or the minimum of the average of the loss function, right? So you do arg max. They are both sort of doing an argmin, right? So in the Bayes Estimator, we wanted to minimize the posterior expected loss, but our Bayes estimator was the value that gave us that minimum posterior expected loss, the argument that minimized it. Here, we're doing the argument that minifies the likelihood function. instead. So they're both, it's the input we care about, not the output. We don't care about the expected loss itself per se. We care about what value gives us that minimum expected loss. We don't care about the likelihood function itself, but we care about what value, what input gives us that minimum.

Any other questions? Yeah. The biggest one that you observed. So when you took probability theory, you talked about order statistics, right?

So if you ordered all of your data, so you rewrote it so that your smallest observed x is on the left and your biggest one is on the right sequential order it's the it's the one on the right side it's just sort of shorthand for the maximum so here i'm saying the maximum of all of these observations x1 through x-n but i might also write max of x i which is just shorthand notation for the maximum of your x's across all the observed eyes so these are me saying the same thing okay i'll stick around for a bit if there's questions Otherwise, have a nice weekend and I will see you on Tuesday.